Quiz: Chubby

Test your knowledge of Chubby by completing this quiz.

Quiz ?

Statement

Chubby is typically deployed as a set of 3-5 servers to provide

Select Multiple Options

Correct
☑ Fault tolerance in case of failure

False
☐ Distributed storage

False
☐ High throughput for read requests

False
☐ High throughput for write requests

Check Answer

Correct Answer ✓

Quiz ?

Statement

The design of Google Chubby can best be described as

Select Multiple Options

False
☐ A set of replicas, each holding a different part of the file system namespace

False
☐ One master with multiple replicas, with client requests load balanced among all live systems

Correct
☑ One active master with multiple replicas as backups that do not process client requests

False
☐ A peer-to-peer distributed hash table

Check Answer

Correct Answer ✓

Quiz ?

Statement

What is Chubby's fault-tolerance model?

Select Multiple Options

False
☐ Active-active

False
☐ Active-passive

Correct
☑ Triple modular redundancy (TMR)

False
☐ Five-way modular redundancy (5-MR)

Check Answer

Correct Answer ✓

Quiz ?

Statement

Chubby clients maintain a constant session with Chubby cell using

Select Multiple Options

Correct
☑ KeepAlive

False
☐ Status

False
☐ StayAlive

False
☐ HeartBeat

Check Answer

Correct Answer ✓

Quiz ?

Statement

Which of the following operation is not possible in Chubby?

Select Multiple Options

Correct
☑ Multiple clients getting a read lock on a file

False
☐ Getting a write locking on a directory

False
☐ Deleting a file or directory

False
☐ Appending to a file

Check Answer

Correct Answer ✓

Quiz ?

Statement

Which of the following statements best describes a client's interaction with Chubby?

Select Multiple Options

False
☐ A client contacts any Chubby server which then forwards the request to the master in the Chubby cell.

False
☐ A client contacts any nearby Chubby server which can process the request since all Chubby servers are replicated.

Correct
☑ A client contacts the Chubby master which handles all client requests.

False
☐ A client contacts the Chubby master which then forwards the request to the appropriate replica in the Chubby cell.

Check Answer

Wrong Answer ✗

Quiz ?

Statement

Which of the following is not a property of Chubby?

Select Multiple Options

False
☐ A distributed lock service that manages leases for resources

False
☐ Uses active replication for fault tolerance

False
☐ Uses Paxos to ensure consistency among servers

Correct
☑ Uses load balancing across all replicas to respond to multiple client requests

Check Answer

Correct Answer ✓

Quiz ?

Statement

What is the use of 'Sequencer' in Chubby?

Select Multiple Options

Correct
☑ To handle receiving messages out of order.

False
☐ To make file handles unique and not guessable.

False
☐ To avoid multiple writers on a file/directory.

False
☐ To uniquely identify cached content.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Chubby does not support

Select Multiple Options

Correct
☑ Partial file reads

Explanation

Chubby was designed to store small files (e.g., configuration data), provide locking, and notifications. Since files are small, file I/O is only entire file reads & writes.

False
☐ Event notifications

False
☐ Whole file uploads

False
☐ File locking

Check Answer

Correct Answer ✓

← Previous
Summary: Chubby

Next →
Mock Interview: Chubby

☐ Mark as Completed